

Wavefield Reconstruction of Teleseismic Receiver Function with the Stretching-and-Squeezing Interpolation Method

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ABSTRACT

A nonlinear stretching-and-squeezing approach is proposed for interpolation and regularization of the teleseismic receiver function (RF) of a regional seismic array. Synthetic experiments for several models show that our method is effective and has the advantages of being able to recover rapid horizontal variations of subsurface interface and mitigate stair-step artifacts. Our method is also applied to a field dataset of the Wabash Valley seismic array. The common-conversion-point (CCP) stacking test shows that high-quality images can be obtained using the interpolated RFs with the nonlinear stretching-and-squeezing interpolation method. Our method can be used as a pragmatic preprocessing tool in RF seismic imaging, migration, and inversion.

INTRODUCTION

Teleseismic receiver functions (RFs) have been routinely used to image Earth's velocity discontinuities (e.g., [Langston, 1979](#); [Crotwell and Owens, 2005](#)). Although single-station methods, such as the H - κ stacking method ([Zhu and Kanamori, 2000](#)), provide valuable information on the local Earth structure beneath a seismic station, modern high-resolution teleseismic waveform or RF imaging techniques utilize the coherent signals of a seismic array to map the subsurface structure (e.g., [Dueker and Sheehan, 1997](#); [Sheehan et al., 2000](#); [Zhu, 2000](#); [Bostock et al., 2001](#); [Morozov and Dueker, 2003](#); [Poppeliers and Pavlis, 2003](#); [Chen et al., 2005](#); [Levander et al., 2005](#); [Wilson and Aster, 2005](#); [Cheng et al., 2016](#); [Hansen and Schmandt, 2017](#)) and require the seismic records to be spatially sampled densely and regularly. However, most passive seismology experiments suffer from sparse, incomplete, and irregular spatial sampling, resulting in distorted images and artifacts (e.g., [Safar, 1985](#); [Chen and Schuster, 1999](#); [Chen et al., 2005](#); [Zhang and Zheng, 2015](#)). The artifacts can be mitigated if appropriate preprocessing interpolation techniques are applied to RFs (e.g., [Zhang and Zheng, 2015](#); [Song et al., 2017](#)). Therefore, development of methods of RF interpolation and regularization is needed.

Various interpolation methods have been proposed in exploration seismology, such as the F - X domain predictions

filter method ([Spitz, 1991](#)), the projection onto convex sets method ([Abma and Kabir, 2006](#)), the nonuniform Fourier transform method ([Duijndam et al., 1999](#)), the wave equation method (e.g., [Canning and Gardner, 1996](#)), the plane-wave destruction filter method ([Fomel, 2000](#)), the curvelet interpolation method (e.g., [Herrmann and Hennenfent, 2008](#)), the matching pursuit method (e.g., [Choi et al., 2016](#)), interferometric interpolation ([Wang et al., 2009](#)), and the nonlinear stretching-and-squeezing method (e.g., [Martinson and Hopper, 1992](#)). Because waveform interpolation is intrinsically an underdetermined problem, proper assumptions regarding the Earth's subsurface geometry need to be provided as additional constraints (e.g., [Spitz, 1991](#)). Most of the methods used in exploration seismology assume evenly distributed observations with a few dead traces and work best for linear events ([Abma and Kabir, 2005](#)). However, teleseismic RFs have several distinguishing characteristics compared to those of exploration seismology data. For instance, RFs are aligned with direct P arrivals with an approximately vertical incidence angle; RFs mainly contain converted phases and their multiples, and the spatial gaps between records are usually irregular and large. Moreover, the signal-to-noise conditions of RFs in a seismic array are usually different, due to the uneven event distribution and the wide aperture of station distribution. Therefore, interpolation methods in exploration seismology are unsuitable for direct application to a passive teleseismic wavefield or RFs.

In addition, three-component teleseismic wavefield reconstruction methods have been proposed, such as the linear Radon transform interpolation ([Wilson and Guitton, 2007](#)), the cubic spline interpolation ([Zhang and Zheng, 2015](#)), the radial basis function interpolation ([Song et al., 2017](#)), and the curvelet transform interpolation ([Shang et al., 2017](#)). After the teleseismic wavefield has been interpolated, RFs are calculated using deconvolution approaches in the time domain ([Ligorria and Ammon, 1999](#)) or the frequency domain ([Langston, 1979](#)). On the other hand, several other approaches aim to perform RF interpolation directly. [Chai et al. \(2015\)](#) introduced a simple weighted summation scheme to interpolate RFs. The interpolated RFs have a high signal-to-noise ratio (SNR), but at the expense of the spatial resolution. [Pavlis](#)

(2011) modified and applied the pseudostation stacking method (Neal and Pavlis, 1999, 2001) of RF interpolation. The pseudostation stacking interpolation method is intrinsically a slant stacking method and has the problem of waveform distortion (Zhang and Zheng, 2015).

In this study, we propose a nonlinear stretching-and-squeezing method to interpolate RFs. Synthetic experiments show that our method can be used to reconstruct RFs in a regional seismic array successfully and has the advantage of mitigating stair-step artifacts. In addition, we apply our method to a real-world dataset of the Wabash Valley seismic array (see Data and Resources). Images resulting from common-conversion-point (CCP) stacking (Zhu, 2000) using original and interpolated RFs are compared, validating the efficacy of our method. Our method can be applied as a useful preprocessing tool for RF seismic imaging, migration, and inversion.

THEORY

In this study, we mainly consider teleseismic P -wave RF as a proof of concept. Assuming that there is one continuous dominant interface (e.g., the Moho) underneath the seismic array, P RFs mainly contain P -to- S converted waves, P_s , of the dominant interface and its free surface multiples, $PpPs$ and $PpSs + PsPs$ (e.g., Ammon, 1991; Zhu and Kanamori, 2000). Because RFs are commonly computed by teleseismic waveforms with epicentral distances in the range of 30° – 100° (e.g., Bostock, 2007), teleseismic P waves in this distance range arrive at subcritical incidence angles, and the relative arrival times of these phases are (Zhu and Kanamori, 2000)

$$t_{P_s} = H \left(\sqrt{\frac{1}{V_s^2} - p^2} - \sqrt{\frac{1}{V_p^2} - p^2} \right), \quad (1)$$

$$t_{PpPs} = H \left(\sqrt{\frac{1}{V_s^2} - p^2} + \sqrt{\frac{1}{V_p^2} - p^2} \right), \quad (2)$$

$$t_{PpSs+PsPs} = 2H \sqrt{\frac{1}{V_s^2} - p^2}, \quad (3)$$

in which H is the depth of the interface, p is the ray parameter, and V_p and V_s are the compression and shear-wave velocity in the crust, respectively. Zhu and Kanamori (2000) showed that the travel times are weakly dependent on V_p but strongly dependent on the V_p/V_s ratio and H . The ray parameter p is approximately constant for a teleseismic event in a local seismic array within a small aperture, in which we assume V_p/V_s to be nearly constant. Therefore,

the travel times t_{P_s} , t_{PpPs} , and $t_{PpSs+PsPs}$ can be linearly related to the local interface depth H .

In this scenario, RFs of a reference station $f^*(t)$ and another arbitrary station $f^m(t)$ within a local seismic array can be decomposed as

$$f^*(t) = \sum_i A^i \delta(t - t_i) \otimes g(t), \quad (4)$$

$$f^m(t) = \sum_i A^i \delta(t - \alpha^m t_i) \otimes g(t), \quad (5)$$

in which $g(t)$ is the source wavelet of RFs, for example, Gaussian functions (Langston, 1979) or the Ricker wavelet (Pavlis and Wang, 2015), and the symbol $\otimes$ denotes the convolution operator. i ranges from 1 to 3, indicating P_s , $PpPs$, and $PpSs + PsPs$ phases, respectively. $\delta(t)$ is the Dirac delta function, indicating a pulse at time t . A^i and t_i represent the amplitude and travel time of the i th phase. α^m is a stretching-and-squeezing coefficient that draws a connection between $f^*(t)$ and $f^m(t)$, which is constant for a particular RF waveform. α^m is independent of i and physically related to the local interface depth ratio H^m/H^* .

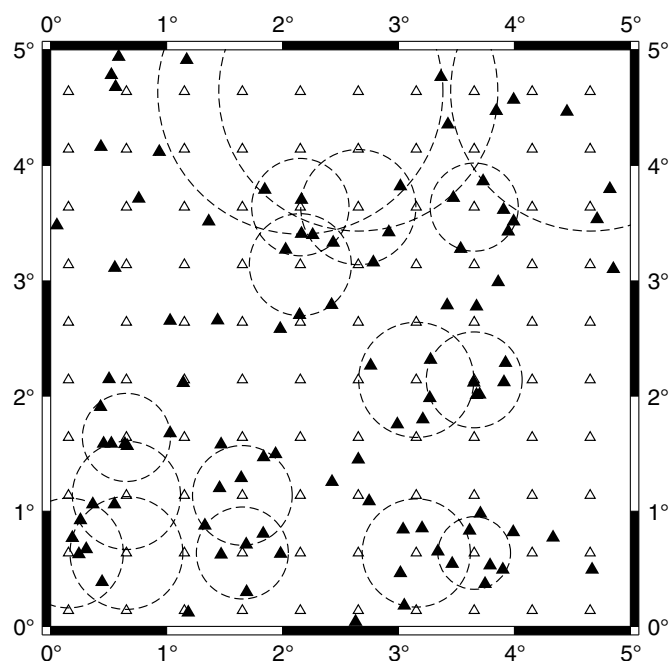
Equations (4) and (5) establish a connection between a reference RF and another arbitrary RF within a local array. To define the criteria to determine whether two RFs are aligned, a discrepancy coefficient $c(\alpha^m)$ is defined by

$$c(\alpha^m) = \|f^*(t) - S_{\alpha^m}[f^m(t)]\|_2, \quad (6)$$

in which $\|\cdot\|_2$ denotes the L2 norm, and S_{α^m} is the nonlinear stretching-and-squeezing operator that is related to stretching-and-squeezing coefficients α^m . For any arbitrary pair of RFs, $c(\alpha^m)$ will approach 0 for appropriate α^m if the two RFs are identical. The best discrepancy coefficient is defined as the minimum $c(\alpha^m)$ within a certain range.

It is noteworthy that the above analysis relies on the assumption that there is one dominant continuous interface and that the V_p/V_s ratio is approximately constant within a local area. In addition, we argue here that it is possible for the method to still work, even if such assumptions are mildly violated. For instance, if other relatively weak additional interfaces exist, or if the V_p/V_s ratio gradually changes within the area, the P_s phase of the dominant interface may exhibit the largest amplitude, and thus, the RFs can still be approximately aligned using equation (6). Below, we validate the method using both synthetic and real data.

The idea of stretching-and-squeezing interpolation was initially proposed by Martinson and Hopper (1992) for reflection exploration seismology. The authors assumed that two seismic traces ($u(x = x_1, t)$ and $u(x = x_2, t)$, in which x_1 and x_2 are locations of receivers) were similar, except that one had undergone a stretching and/or squeezing of its time axis relative to that of the other. The mapping function was

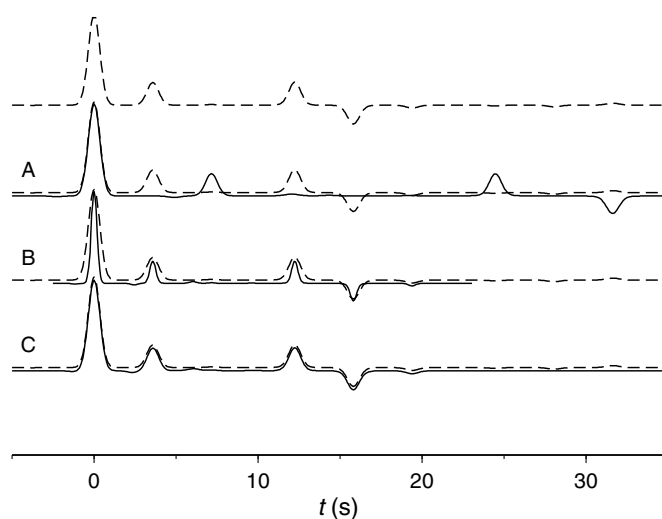


▲ **Figure 1.** A sketch showing the determination of the aperture centered at each virtual station, within which receiver functions (RFs) are interpolated. Open triangles denote the locations of virtual stations. Solid triangles denote the locations of recorded RFs. Dashed lines denote the radius of aperture within which there are at least n recorded RFs (here, we choose $n = 6$).

parameterized as a truncated Fourier sine series with unknown coefficients (Martinson and Hopper, 1992), which is rather complex to compute. The mapping function for RFs is much simpler, as shown in equation (5) and can be approximately described using only one parameter, that is, the stretching-and-squeezing coefficient α^m .

IMPLEMENTATIONS

Equations (4) and (5) will be better satisfied for RFs spatially closer to each other. Accordingly, the nonlinear stretching-and-squeezing method should be performed within a sufficiently small area. We propose the algorithm sketched in Figure 1. First, the appropriate radius of aperture centered at each virtual station is determined. Within each aperture, there is a certain number of recorded RFs, and equations (4) and (5) are satisfied. Then, an arbitrary set of recorded RFs within the aperture is chosen as a reference, and the best coefficients for the other recorded RFs within the same aperture can be obtained according to equation (6). Our synthetic tests show that the final interpolation results will not be considerably influenced by the choice of a particular recorded RF as a reference. However, the choice of reference RF may be important for real-world data, partially because the SNR varies from trace to trace. In principle, if a recorded RF has a high SNR, this recorded RF is suitable for being chosen as a reference.

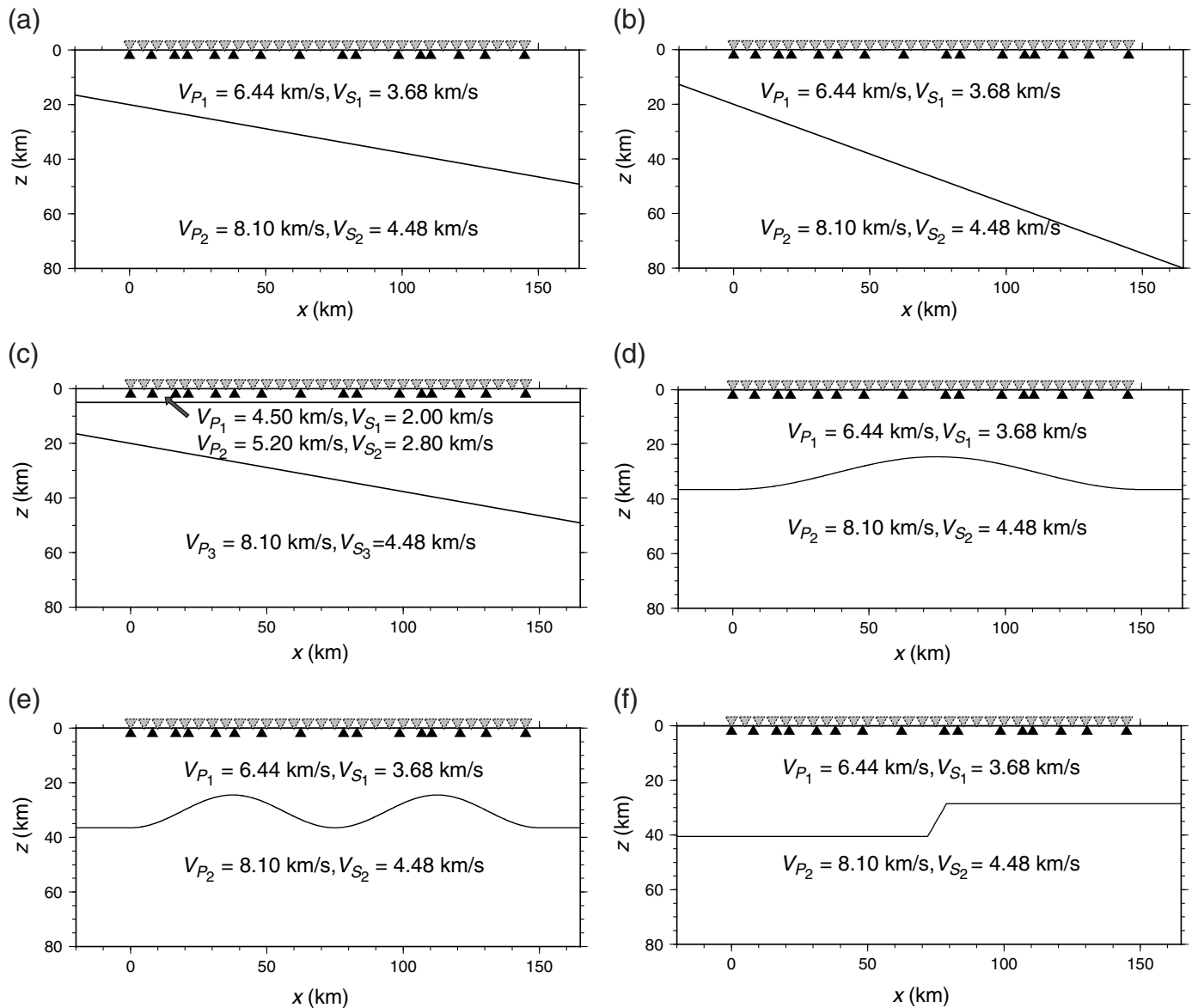


▲ **Figure 2.** Implementation of the stretching-and-squeezing operator S_{α^m} . Dashed lines denote the reference RF $f^*(t)$. Solid lines denote the RF that the operator S_{α^m} acts on. A denotes comparison of reference RF $f^*(t)$ to another RF $f^m(t)$ within one aperture. B denotes simply stretching or squeezing the RF $f^m(t)$. C denotes restoring the shape of the Gaussian wavelet of $f^m(t)$.

Once all α^m within one aperture have been obtained, all RFs within that aperture can be aligned using the nonlinear operation S_{α^m} . The RF waveform of the virtual stations can then be obtained using the cubic spline interpolation (Zhang and Zheng, 2015), radial basis function interpolation (Song et al., 2017), or the pseudostation stacking method (Neal and Pavlis, 1999, 2001). At the same time, the stretching-and-squeezing coefficients of virtual stations α^m can also be interpolated simultaneously. Finally, the nonlinear inverse operation $S_{\alpha^m}^{-1}$ can be applied to the interpolated waveforms, and the final RF interpolation results can be obtained. Using the procedure discussed above, RFs of all virtual stations can be obtained one by one.

The key step to carry out the nonlinear interpolation is to implement the stretching-and-squeezing operation S_{α^m} . At first glance, for the reference RF and another RF within the same aperture (Fig. 2a), it appears that it can be implemented simply by stretching or squeezing the recorded RF and then comparing it to the reference. However, the simple stretching-and-squeezing operation will distort the wavelet (Fig. 2b). An additional operation must also be performed to restore the shape of the wavelet (Fig. 2c). Here, we implement the nonlinear operator by using the time-domain iterative deconvolution (Ligorria and Ammon, 1999) to deconvolve the distorted wavelet pulses and then convolve with the wavelet of the reference (Fig. 2c). The inverse stretching-and-squeezing operation $S_{\alpha^m}^{-1}$ is implemented in a similar manner.

After the mapping function α^m within one aperture has been obtained, all RF waveforms within the aperture can be aligned, and virtual station stacks can be obtained. Martinson and Hopper (1992) proposed a linear scheme for interpolation



▲ Figure 3. Configuration of six models for synthetic tests. (a) Crustal models with the Moho dipping 10° and (b) 20° in the x -axis direction; (c) a two-layer model with a middle horizontal crust on the top and the Moho dipping 10° ; (d) a sinuous Moho, smoothly varying in the x -axis direction with the wavelength of 150 km; (e) a sinuous Moho with the reduced wavelength of 75 km; (f) a crust model, in which the Moho is offset by 12 km over a horizontal distance of 7 km. Station distributions of recorded RFs (black triangles) and virtual stations (gray triangles) are shown at the surface of each model. The real stations are randomly distributed, and virtual stations are distributed with a station spacing of 5 km. Velocities of each block of the models are labeled in each figure.

that yields unfavorable discontinuous derivatives (Zhang and Zheng, 2015). In this study, we use the cubic spline scheme (Zhang and Zheng, 2015) to interpolate the wavefield, which offers smoother results (Zhang and Zheng, 2015), and no subjective parameters are used. To search for the best stretching-and-squeezing parameter, we use the grid-search method to find the global minimum in a certain range. Because the stretching-and-squeezing parameter α^m has a clear physical meaning (the ratio of local interface depths H^m/H^*), a reasonable search range can be provided according to preliminary results on the regional Moho depth variation. For instance, the search range

can be smaller for areas with a slowly varying Moho. In areas where previous studies are unavailable, the search range can be determined by visually examining the Moho P_s arrival-time variation for different stations and by trial and error.

SYNTHETIC EXPERIMENTS

To validate the efficacy of our method, we calculate synthetic RFs for the following six models: two crustal models with the Moho dipping 10° (Fig. 3a) and 20° (Fig. 3b) in the x -axis direction, a two-layer model with a middle horizontal crust

on the top and the Moho dipping 10° (Fig. 3c), a sinuous Moho, smoothly varying in the x -axis direction with the wavelength of 150 km (Fig. 3d), a sinuous Moho with a reduced wavelength of 75 km (Fig. 3e), and a crustal model in which the Moho is offset by 12 km over a horizontal distance of 7 km (Fig. 3f). Then, we perform synthetic tests using our nonlinear interpolation method with the calculated RFs. For all of these synthetic tests, the RFs of the 16 real stations are interpolated onto 30 evenly distributed virtual stations with a station spacing of 5 km ranging from 0 to 145 km (Fig. 3). Details of model parameters, the locations of the real stations and of the virtual stations are shown in Figure 3.

For the dipping-layered models, the Z and R components of teleseismic waveforms are calculated using the anisotropic ray theory code (RAYSUM) package (Frederiksen and Bostock, 2000) that can be used to efficiently obtain the high-frequency asymptotic teleseismic wavefield. For the sinuous-Moho and step-Moho models, the teleseismic waveforms are calculated using the hybrid method that interfaces the spectral element method (SEM) with the frequency-wavenumber (FK) method (SEM-FK) method (Tong *et al.*, 2014). After the teleseismic waveforms have been obtained, all RFs are computed by the time-domain iterative deconvolution method (Ligorria and Ammon, 1999), with a low-pass Gaussian filter ($a = 3$) applied as the wavelet of RFs.

In the first example, we compare the modeled and interpolated RFs for dipping-layered models (Fig. 3a,b) to validate the correctness and advantage of our method. The results show that the RFs waveforms can be perfectly recovered using our nonlinear interpolation method (Fig. 4). We also compare the interpolated RFs using our method and using the method of Zhang and Zheng (2015) (Fig. 4). The results show that both methods can successfully reconstruct the P_s -converted phases. However, our method has the advantage of mitigating stair-step artifacts of multiples, especially for phases with a large dipping angle (Fig. 4).

In the second example, to test the performance of our method applied to noisy data, we add Gaussian noise to the teleseismic waveforms of dipping-layered models (Fig. 3a,b) for synthetic tests. Original and interpolated RFs for each model are shown in Figure 5. Results show that, although RF waveforms can no longer be exactly recovered, the P_s -converted phase and its multiples can still be successfully reconstructed. This test indicates that our method can properly interpolate and regularize RFs even in the presence of noise. In this test, we add random Gaussian noise to the teleseismic waveforms prior to deconvolution using MATLAB's (see Data and Resources) built-in function Add White Gaussian Noise (AWGN). Gaussian noise is added to every time sampling point independently for the three-component waveforms at the SNR of 20. It is noteworthy that the features of real-world data are quite complex, because the noise in vertical and horizontal components of RFs may be correlated and bandlimited. Here, we do not consider such effects. The real RFs with the same SNR may be as clean as shown in our test.

In the third example, we test our method in a more complex scenario for models in Figure 3c,d. For the two-layer-over-half-space model in Figure 3c, there are two P_s phases, whereas for the sinuous-Moho model in Figure 3d, the P_s phase travel time can no longer be exactly described by equation (1). Interestingly, results show that the RF waveforms can still be perfectly reconstructed for these models using our method (Fig. 6). This test indicates that the relationship between adjacent RFs in these models can still be approximately described by the stretching-and-squeezing coefficient, and our method even works when the underlying assumption is mildly violated. It is noteworthy that we process the RF interpolation exactly as for single dipping-layered models, and no additional operation is needed.

In the fourth example, we test our method in the case of a strong Moho topography variation (Fig. 3e,f). Synthetic RFs and interpolated RFs are shown in Figure 7. The tests show that our methods can reconstruct most of RF waveforms successfully, even for the model with a strong lateral variation. It is noteworthy that near the Moho offset, RF waveforms cannot be recovered perfectly (e.g., V015 for the step-Moho model). This result is probably due to the existence of a strong diffraction wave from the corner in this trace. This test indicates that our method can work for some extreme cases, but some traces with strong diffraction waves from the corners cannot be fully recovered.

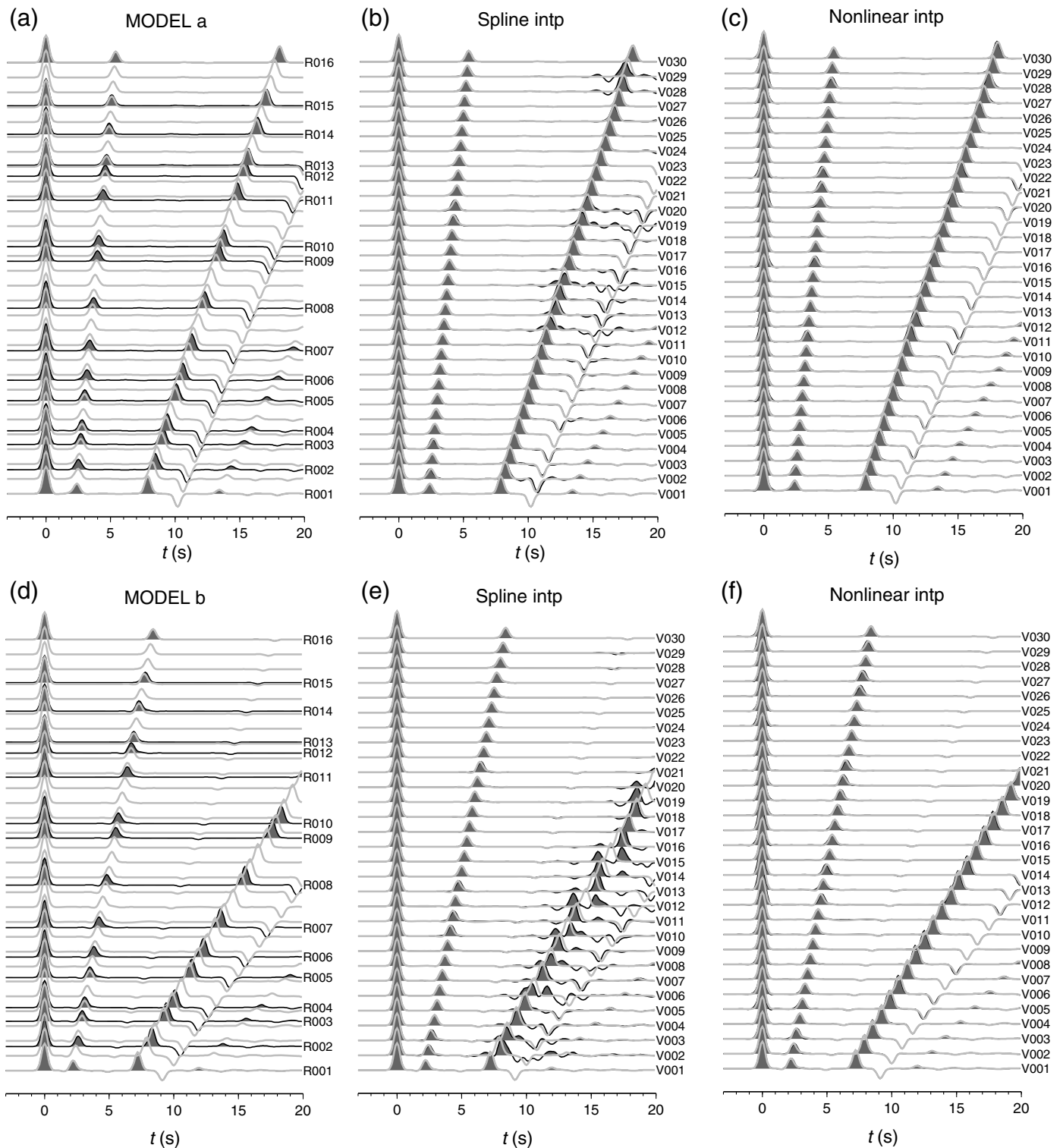
According to the above synthetic tests, our nonlinear interpolation method can reconstruct RF waveforms successfully in most cases, especially for models with a mild horizontal variation. Our method works well in most parts of the waveform, even for the models with a large Moho topography variation; however, strong diffraction waves from the corners cannot be fully addressed.

APPLICATION TO FIELD DATASETS

Next, we apply our nonlinear stretching-and-squeezing interpolation method to the real-world RF datasets of the Wabash Valley seismic array in the central United States (Fig. 8).

The Wabash Valley area is located in the adjacent regions of Illinois, Indiana, Missouri, and Kentucky (Fig. 8). The Wabash Valley seismic zone seismic experiment was jointly conducted by researchers from Saint Louis University and Southern Illinois University during 2014–2016 (see Data and Resources). A seismic array with 47 broadband stations was deployed in the Wabash Valley in Illinois and Indiana, with an average station spacing of 5 km. The RFs were calculated using a time-domain iterative deconvolution approach (Ligorria and Ammon, 1999), with a low-pass Gaussian filter ($a = 3$), and data were manually selected for quality control (see Data and Resources).

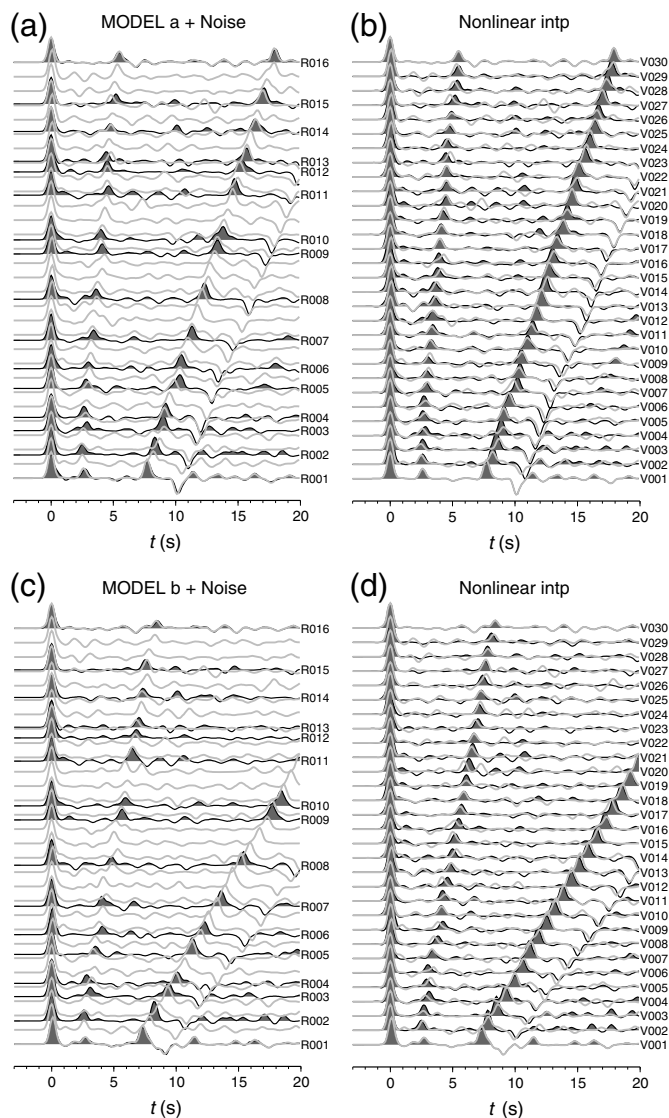
An example of RFs of this seismic array for an event and the corresponding interpolated RFs using the nonlinear interpolation method are shown in Figure 9. Several interesting features can be observed in the interpolated RF profile. For instance, a continuous Moho P_s -converted phase is clearly observed at approximately 7 s; it is not as apparent in the original RF records at both ends of the cross section where the



▲ **Figure 4.** Synthetic RFs for models in (a) Figure 3a and (d) Figure 3b, respectively; (b,e) the corresponding cubic spline interpolation results; and (c,f) nonlinear stretching-and-squeezing interpolation results. In (a,d), black lines represent the randomly distributed recorded RFs. Gray lines represent ground-true RFs. All traces were organized by station locations, and station names were labeled next to each trace.

distribution of stations is relatively sparse. The interpolated RFs between WB05 and WB07 traces appear to be reasonable, because the P_s phase at approximately 7 s and their multiples at approximately 15 s are coherent.

We further apply the CCP stacking method implemented by Zhu (2000) to the interpolated RFs (Fig. 10) and compare it to the CCP result for the original RFs (see Data and Resources). The comparison shows that the CCP image quality is improved

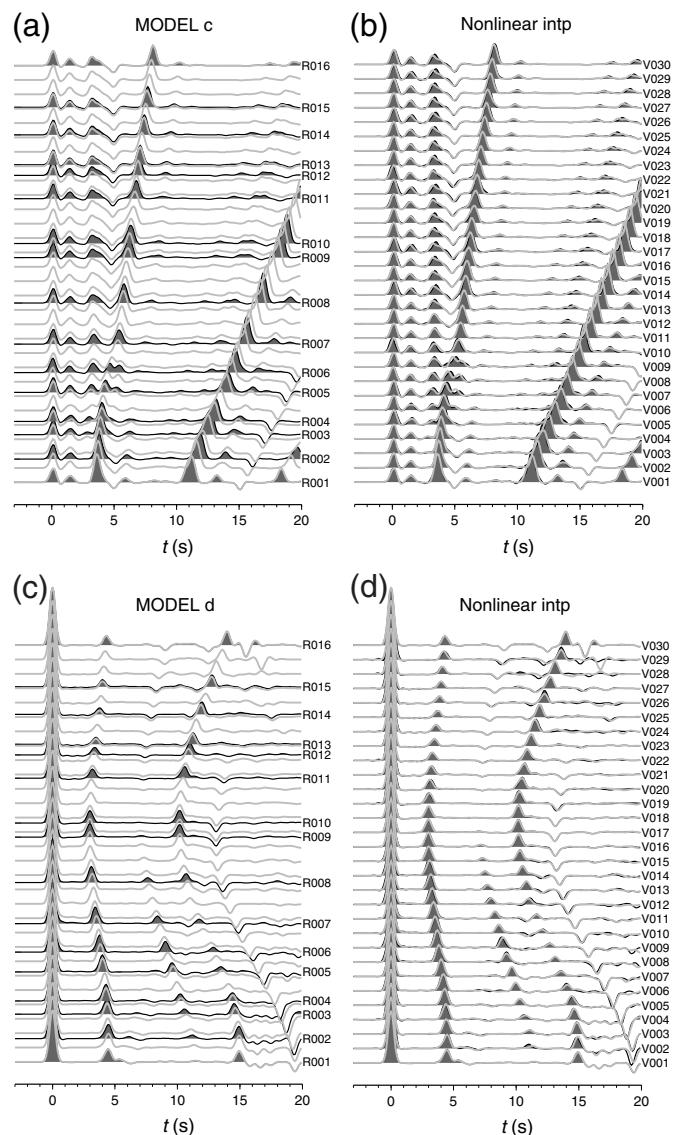


▲ **Figure 5.** Synthetic RFs for models in (a) Figure 3a and (c) Figure 3b with noise; and (b,d) the corresponding interpolation results using the nonlinear stretching-and-squeezing interpolation method. Random Gaussian noise was added to the teleseismic waveforms prior to deconvolution. See caption of Figure 4 for details of the labels.

with interpolated RFs. For instance, the CCP result with interpolated RFs is relatively smooth, and the missing gaps at both ends of the cross sections are filled. In addition, the original negative anomaly at the approximate horizontal distance of 100 km and depth of 20 km is no longer seen in the CCP image with interpolated RFs. This observation implies that the anomaly in the original CCP image may be an artifact.

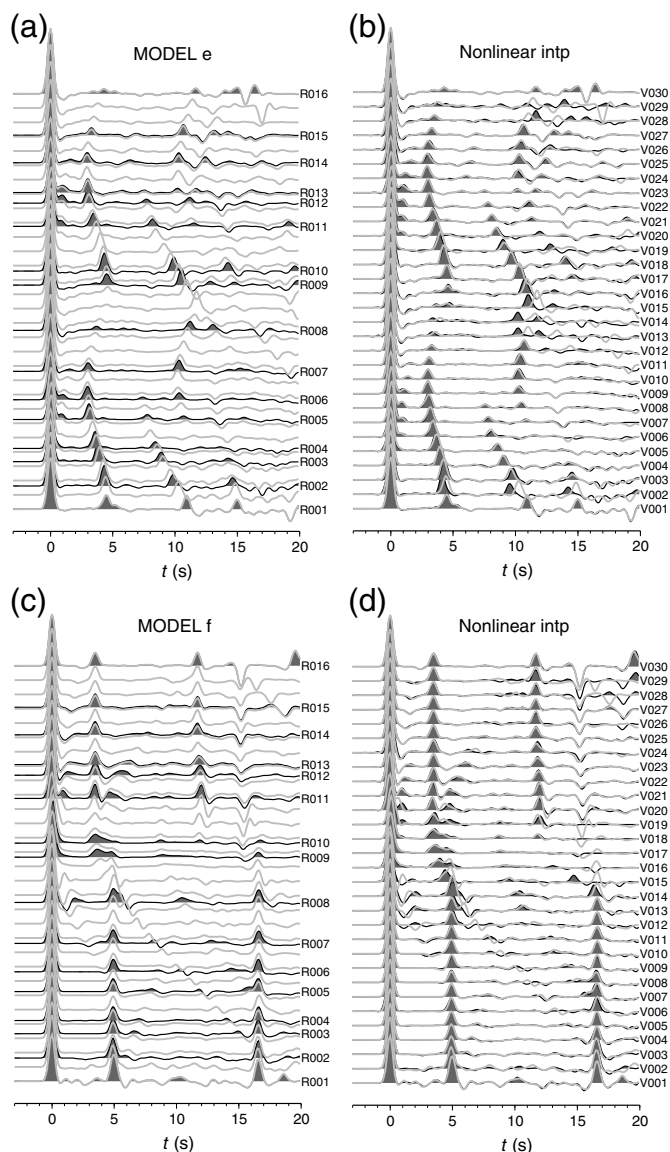
DISCUSSION AND CONCLUSIONS

The choice of aperture size for RF interpolation is not trivial. If inappropriate interpolation aperture size is used, spatial aliasing problems may occur. Because the resolution of the images is



▲ **Figure 6.** Synthetic RFs for models in (a) Figure 3c and (c) Figure 3d; and (b,d) the corresponding interpolation results using the nonlinear stretching-and-squeezing interpolation method. See caption of Figure 4 for details of the labels.

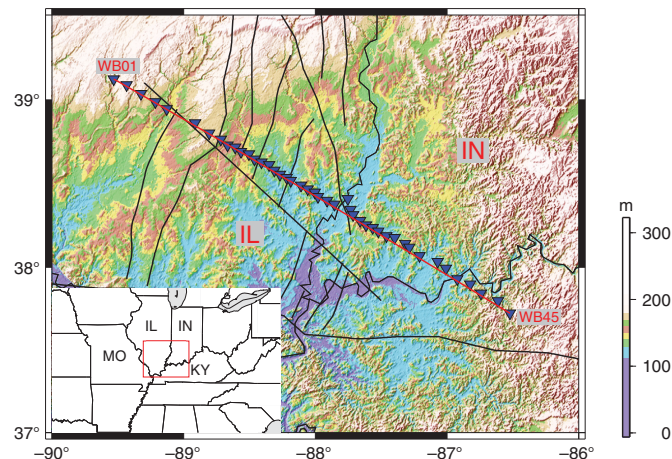
determined by the size of the Fresnel zone, Pavlis (2011) uses a time-dependent Fresnel zone variable to control the aperture size for RF interpolation. In this study, the choice of the aperture size depends on the validation of the assumption that the main interface (e.g., the Moho) between two nearby stations is smooth and continuous. The correctness of the assumption is usually restricted to a local area and also depends on how rapid the lateral variation of the earth structure is. In our synthetic tests, the missing RF waveforms can be perfectly reconstructed for the RFs of dipping-layered models (Fig. 3a,b) and for the slowing varying model (Fig. 3d). However, strong diffraction waves from the corners for the models with a strong lateral variation (Fig. 3e,f) cannot be successfully recovered. These tests suggest that the appropriate aperture size depends on the



▲ **Figure 7.** Synthetic RFs for models in (a) Figure 3e and (c) Figure 3f; and (b,d) the corresponding interpolation results using the nonlinear stretching-and-squeezing interpolation method. See caption of Figure 4 for details of the labels.

specific model that we address. For instance, for the dipping-layered models and slowly varying models, the aperture size can be relatively large, whereas the aperture size for models with a strong lateral variation should be relatively small. Because choosing the aperture size is always complex in real-world cases, and we are usually unsure about the underlying structure, a reasonable strategy may be to choose the aperture size of 2–3 Fresnel zones as suggested by Pavlis (2011).

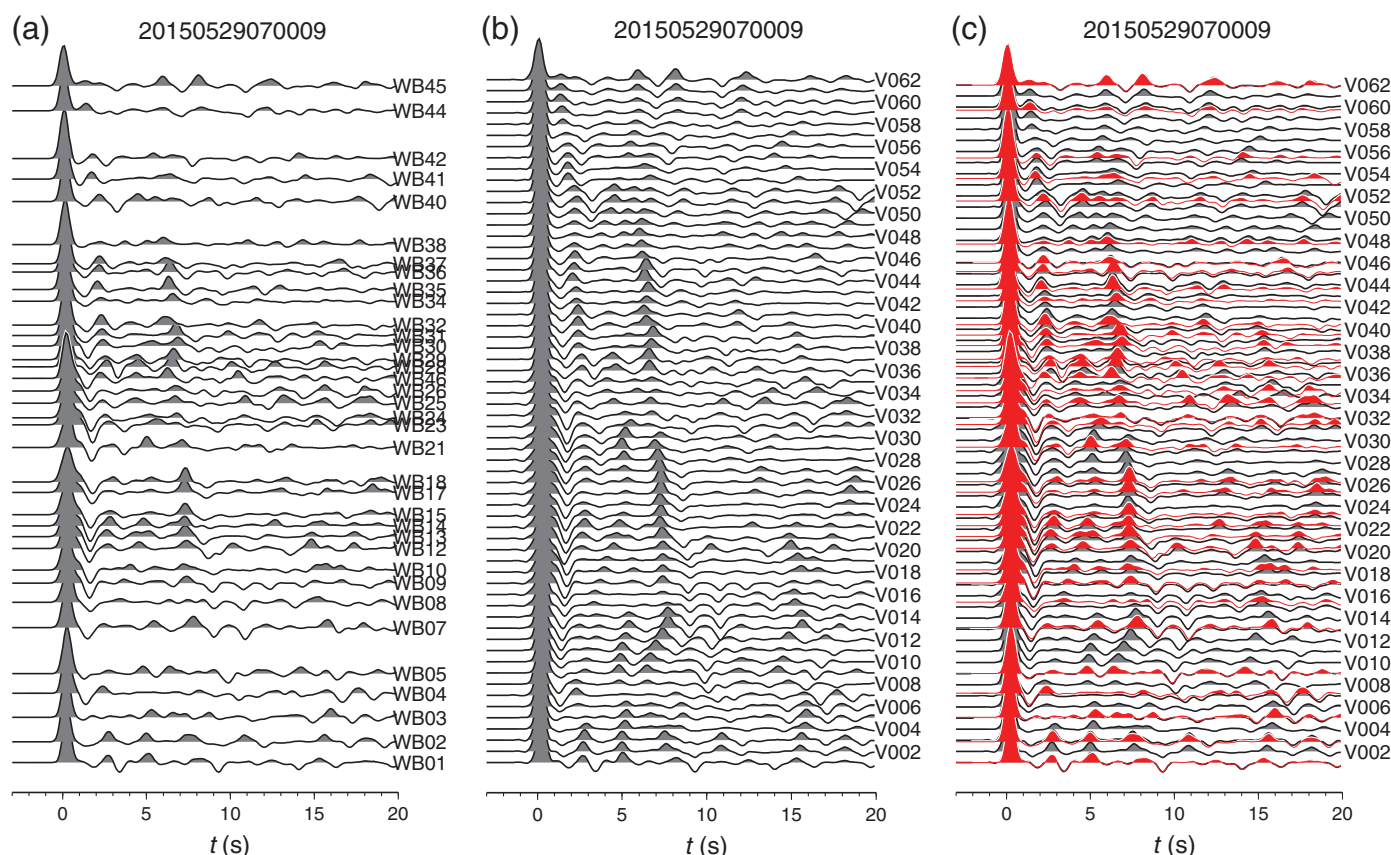
Generally, there are two approaches to obtaining interpolated RFs from raw three-component teleseismic waveforms: the first, called the preprocessing approach, is to reconstruct the wavefield and then calculate RFs; the other approach, described as the postprocessing approach (Song *et al.*, 2017),



▲ **Figure 8.** Topographic maps and station distribution of the Wabash Valley area in the central United States. Blue triangles denote the distribution of stations. Thick black lines denote major regional faults. The red line denotes the cross section along which RFs are displayed and interpolated in Figure 9. (Inset) Black lines denote state boundaries. The red rectangle shows the location of the Wabash Valley area. MO, Missouri; IL, Illinois; IN, Indiana; KY, Kentucky.

is to calculate RFs first and then perform the interpolation. (Song *et al.*, 2017) considered the step-Moho model and performed numerical tests to investigate the interpolated RF results using both approaches. The authors concluded that the preprocessing approach appears to perform better. Our nonlinear interpolation method is a kind of postprocessing method. Based on our numerical tests, it appears that which approach performs better is case dependent. For instance, for the dipping-Moho model tested in the preceding sections (Figs. 4 and 5), the postprocessing approach indeed provides reasonable results. However, the approach is usually much more complex for real-world data, partially because the noise level may vary significantly from station to station, especially when automatic quality control is adopted for preprocessing (Crotwell and Owens, 2005; Pavlis, 2011). In such a case, preprocessing may be intrinsically more robust. Therefore, care must be taken when using the postprocessing approach, especially in RF studies for a large area, such as the USArray. In certain regional experiments, such as the Wabash experiment, the RFs are preprocessed manually for quality control, and the SNR is high. Our method will be especially useful in these cases.

The success of our nonlinear stretching-and-squeezing interpolation method relies on the validity of assumptions that there is one dominant continuous interface and that the V_P/V_S ratio is constant. These assumptions have typically been valid in most previous RF studies (e.g., Zandt and Ammon, 1995; Zhu and Kanamori, 2000; Gilbert *et al.*, 2003; Fee and Dueker, 2004; Bostock, 2007). Furthermore, we try to satisfy these assumptions better by limiting the usage of our method within a small aperture. Generally, a larger aperture will result in a smoother result, whereas a smaller aperture is better for models with rough Moho



▲ **Figure 9.** (a) Example of recorded RFs; (b) the corresponding interpolated RFs; and (c) the original and corresponding interpolated RFs overlaid for comparison of the Wabash Valley Seismic Experiment. Station names are labeled next to each trace. The cross section is along the thick red line in Figure 8. (c) Black lines are the interpolated RFs, and red lines are the recorded RFs.

variations. Synthetic experiments in the preceding sections show that our method works well, even if the assumptions are not strictly satisfied (Figs. 6 and 7). Under certain extreme conditions when the assumptions are violated, other complementary preprocessing steps need to be applied. For instance, a recent study has demonstrated the possibility of producing phase-specific P_s RFs (Ainiwaer and Gurrola, 2017). Our method can be used to interpolate the phase-specific P_s RFs instead of the original RFs. It is noteworthy that because interpolation is intrinsically an underdetermined problem, some artifacts are likely to be introduced to the RF images when an interpolation method is applied without taking due care.

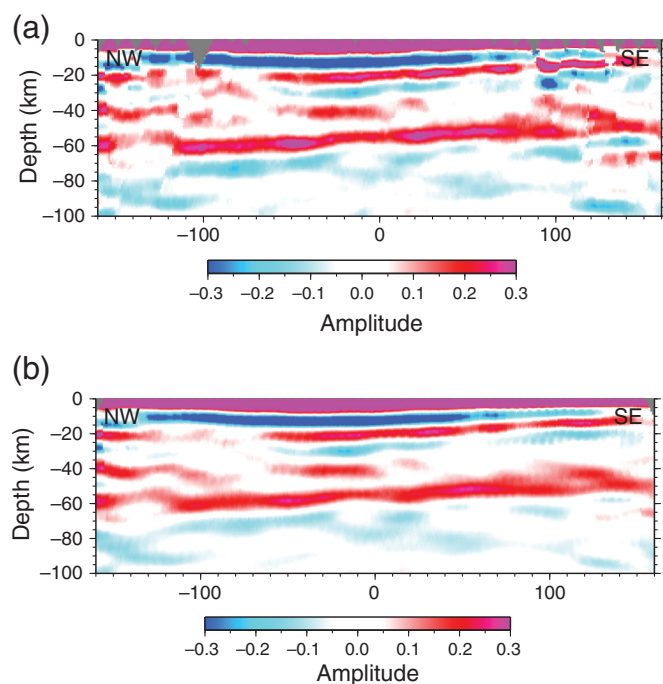
In summary, we propose a nonlinear stretching-and-squeezing RF interpolation method. Synthetic experiments show that this method is able to properly interpolate and regularize unevenly distributed RFs for dipping-layered models, even in the presence of large noise. Our method has the advantage of being able to mitigate stair-step artifacts. Applications of our method to a real dataset further demonstrate the method's validity, showing that the nonlinear interpolation method helps obtain high-quality images. Our method can be used as a preprocessing step in RF seismic imaging, migration, and inversion.

DATA AND RESOURCES

The RAYSUM package is provided by A. Frederiksen of the University of Manitoba (<https://home.cc.umanitoba.ca/~frederik/Software/>, last accessed July 2018). The SEM-FK package is provided by Ping Tong of Nanyang Technological University, Singapore (Tong *et al.*, 2014). The Wabash Valley seismic experiment was jointly conducted by researchers from Saint Louis University and Southern Illinois University during 2014–2016 (<http://www.eas.slu.edu/People/LZhu/research/WabashValley/project.html>, last accessed July 2018), receiver functions (RFs) were collected and preprocessed by Ms Zanjani from Saint Louis University. CCP image of the original RFs was also obtained by Ms Zanjani. All figures are plotted using Generic Mapping Tools v. 4.5 (Wessel *et al.*, 2013). MATLAB is available at (www.mathworks.com/products/matlab, last accessed July 2018) ☒

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▲ **Figure 10.** (a) Comparison of common-conversion-point (CCP) results with original RFs and (b) interpolated RFs in the Wabash Valley seismic experiment. The cross section is along the thick red line in Figure 8 from northwest to southeast.

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REFERENCES

- Abma, R., and N. Kabir (2005). Comparisons of interpolation methods, *The Leading Edge* **24**, 984–989.
- Abma, R., and N. Kabir (2006). 3D interpolation of irregular data with a POCS algorithm, *Geophysics* **71**, E91–E97.
- Ainiwaer, A., and H. Gurrola (2017). Wavefield iterative deconvolution to remove multiples and produce phase specific P_s receiver functions, *Geophys. J. Int.* **212**, 2016–2030.
- Ammon, C. J. (1991). The isolation of receiver effects from teleseismic P waveforms, *Bull. Seismol. Soc. Am.* **81**, 2504–2510.
- Bostock, M. G. (2007). Teleseismic body-wave scattering and receiver-side structure, in *Treatise on Geophysics*, G. Schubert (Editor), Vol. 1, 219–246, Elsevier, Amsterdam, The Netherlands.
- Bostock, M. G., S. Rondenay, and J. Shragge (2001). Multiparameter two-dimensional inversion of scattered teleseismic body waves 1. Theory for oblique incidence, *J. Geophys. Res.* **106**, no. 12, 30,771–30,782.
- Canning, A., and G. H. F. Gardner (1996). Reducing 3-D acquisition footprint for 3-D DMO and 3-D prestack migration, *Geophysics* **63**, 1177–1183.
- Chai, C., C. J. Ammon, M. Maceira, and R. B. Herrmann (2015). Inverting interpolated receiver functions with surface wave dispersion and gravity: Application to the western U.S. and adjacent Canada and Mexico, *Geophys. Res. Lett.* **42**, 4359–4366.
- Chen, J., and G. T. Schuster (1999). Resolution limits of migrated images, *Geophysics* **64**, no. 4, 1046–1053.

- Chen, L., L. Wen, and T. Zheng (2005). A wave equation migration method for receiver function imaging: 1. Theory, *J. Geophys. Res.* **110**, no. B11309, doi: [10.1029/2005JB003665](https://doi.org/10.1029/2005JB003665).
- Cheng, C., T. Bodin, and R. M. Allen (2016). Three-dimensional pre-stack depth migration of receiver functions with the fast marching method: A Kirchhoff approach, *Geophys. J. Int.* **205**, 819–829.
- Choi, J., J. Byun, S. J. Seol, and Y. Kim (2016). Wavelet-based multi-component matching pursuit trace interpolation, *Geophys. J. Int.* **206**, 1831–1846.
- Crotwell, H. P., and T. J. Owens (2005). Automated receiver function processing, *Seismol. Res. Lett.* **276**, 702–708.
- Dueker, K. G., and A. F. Sheehan (1997). Mantle discontinuity structure from midpoint stacks of converted P to S waves across the Yellowstone hotspot track, *J. Geophys. Res.* **102**, no. B4, 8313–8327.
- Duijndam, A. J. W., M. A. Schonewille, and C. O. H. Hindriks (1999). Reconstruction of band-limited signals, irregularly sampled along one spatial direction, *Geophysics* **64**, 524–538.
- Fee, D., and K. Dueker (2004). Mantle transition zone topography and structure beneath the Yellowstone hotspot, *Geophys. Res. Lett.* **31**, L18603, doi: [10.1029/2004GL020636](https://doi.org/10.1029/2004GL020636).
- Fomel, S. (2000). Application of plane-wave destruction filters, *Geophysics* **67**, no. 6, 1946–1960.
- Frederiksen, A. W., and M. G. Bostock (2000). Modelling teleseismic waves in dipping anisotropic structures, *Geophys. J. Int.* **141**, 401–412.
- Gilbert, H. J., A. F. Sheehan, K. G. Dueker, and P. Molnar (2003). Receiver functions in the western United States, with implications for upper mantle structure and dynamics, *J. Geophys. Res.* **108**, no. B5, 2229.
- Hansen, S. M., and B. Schmandt (2017). P and S wave receiver function imaging of subduction with scattering kernels, *Geochem. Geophys. Geosyst.* **18**, no. 12, 4487–4502.
- Herrmann, F. J., and G. Hennenfent (2008). Non-parametric seismic data recovery with curvelet frames, *Geophys. J. Int.* **173**, 233–248.
- Langston, C. A. (1979). Structure under Mount Rainier, Washington, inferred from teleseismic body waves, *J. Geophys. Res.* **84**, 4749–4762.
- Levander, A., F. Niu, and C. T. A. Lee (2005). Imaging teleseismic P to S scattered waves using the Kirchhoff integral, in *Array Analysis of Broadband Seismograms*, A. Levander and G. Nolet (Editors), American Geophysical Union, Washington D.C., 149–295.
- Ligorria, J. P., and C. J. Ammon (1999). Iterative deconvolution and receiver-function estimation, *Bull. Seismol. Soc. Am.* **85**, 1395–1400.
- Martinson, D. G., and J. R. Hopper (1992). Nonlinear seismic trace interpolation, *Geophysics* **57**, 136–145.
- Morozov, I. B., and K. G. Dueker (2003). Depth-domain processing of teleseismic receiver functions and generalized three-dimensional imaging, *Bull. Seismol. Soc. Am.* **93**, no. 5, 1984–1993.
- Neal, S. L., and G. L. Pavlis (1999). Imaging P -to- S conversions with multichannel receiver functions, *Geophys. Res. Lett.* **26**, 2581–2584.
- Neal, S. L., and G. L. Pavlis (2001). Imaging P -to- S conversions with broad-band seismic arrays using multichannel time-domain deconvolution, *Geophys. J. Int.* **147**, 57–67.
- Pavlis, G. L. (2011). Three-dimensional, wavefield imaging of broadband seismic array data, *Comput. Geosci.* **37**, 1054–1066.
- Pavlis, G. L., and Y. Wang (2015). Shaping wavelet effects in scattered wave imaging of P to S conversion data, *Geophys. J. Int.* **203**, no. 1, 373–383.
- Poppeliers, C., and G. L. Pavlis (2003). Three-dimensional, prestack, plane wave migration of teleseismic P -to- S converted phases: 1. Theory, *J. Geophys. Res.* **108**, 2112.
- Safar, M. H. (1985). On the lateral resolution achieved by Kirchhoff migration, *Geophysics* **50**, no. 7, 1091–1099.
- Shang, X., M. V. de Hoop, and R. D. van der Hilst (2017). Common conversion point stacking of receiver functions versus passive-source reverse time migration and wavefield regularization, *Geophys. J. Int.* **209**, 923–934.

Sheehan, A. F., P. M. Shearer, H. J. Gilbert, and K. G. Duckert (2000). Seismic migration processing of *P-SV* converted phases for mantle discontinuity structure beneath the Snake River Plain, western United States, *J. Geophys. Res.* **105**, 19,055–19,065.

Song, P., X. Zhang, Y. Liu, and J. Teng (2017). Moho imaging based on receiver function analysis with teleseismic wavefield reconstruction: Application to South China, *Tectonophysics* **718**, 118–131.

Spitz, S. (1991). Seismic trace interpolation in the *F-X* domain, *Geophysics* **56**, 785–794.

Tong, P., C. Chen, D. Komatitsch, P. Basini, and Q. Liu (2014). High-resolution seismic array imaging based on an SEM-FK hybrid method, *Geophys. J. Int.* **197**, 369–395.

Wang, Y., Y. Luo, and G. T. Schuster (2009). Interferometric interpolation of missing seismic data, *Geophysics* **74**, no. 3, S137–S145.

Wessel, P., W. H. F. Smith, R. Scharroo, J. F. Luis, and F. Wobbe (2013). Generic Mapping Tools: Improved version released, *Eos Trans. AGU* **94**, 409–410.

Wilson, C., and A. Guitton (2007). Teleseismic wavefield interpolation and signal extraction using high-resolution linear radon transforms, *Geophys. J. Int.* **168**, 171–181.

Wilson, D., and R. Aster (2005). Seismic imaging of the crust and upper mantle using regularized joint receiver functions, frequency-wave number filtering, and multimode Kirchhoff migration, *J. Geophys. Res.* **110**, no. B05305, doi: [10.1029/2004JB003430](https://doi.org/10.1029/2004JB003430).

Zandt, G., and C. J. Ammon (1995). Continental crust composition constrained by measurements of crustal Poisson's ratio, *Nature* **374**, no. 9, 152–153.

Zhang, J., and T. Zheng (2015). Receiver function imaging with reconstructed wavefields from sparsely scattered stations, *Seismol. Res. Lett.* **86**, 165–172.

Zhu, L. (2000). Crustal structure across the San Andreas fault, southern California from teleseismic converted waves, *Earth Planet. Sci. Lett.* **179**, 183–190.

Zhu, L., and H. Kanamori (2000). Moho depth variation in southern California from teleseismic receiver functions, *J. Geophys. Res.* **105**, 2969–2980.

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